

In this Standard the comma has been adopted as the decimal sign

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1. SCOPE

This Standard establishes the requirements for chemical milling of series 2000, 6000, and 7000 aluminum alloy parts.

2. APPLICABLE DOCUMENTS

Embraer Standards

- doc.emb 005901 - “Padrão de Processo para Usinagem Química”.
- MEP 13-007 - Sulfuric Acid, Technical.
- MEP 13-008 - Sodium Hydroxid, Technical Grade (Caustic Soda).
- MEP 13-012 - Nitric Acid, Technical.
- MEP 13-013 - Liquid Deoxidizer for Aluminum.
- MEP 13-032 - Triethanolamine (TEA).
- MEP 13-073 - Organic Solvent for Cleaning.
- MEP 13-074 - Masking Compound for Chemical Milling.
- MEP 13-087 - Additive for Chemical Milling Solution.
- MEP 13-091 - Defoamer Additive for Chemical Milling Process.
- MEP 13-097 - Organic Solvent Free From Chlorinated Hydrocarbons Multipurpose.
- MEP 13-118 - Chemical Milling Mask 2K
- MEP 21-017 - Nonsilicated, Nonchromated Alkaline Degreasing Compound.
- NE 03-004 - Surface Roughness.
- NE 40-077 - Equipment for Thermal Processing - Temperature Control and Classification.
- NE 41-012 - Aluminum and its Alloys - Protection and Preparation for Chemical Milling.

NOTE

The doc.emb procedure is for Embraer use only.

Foreign Standards

ASTM - American Society for Testing and Material (USA)

D4376 - Standard Specification for Vapor-Degreasing Grade Perchloroethyle.

Federal Specification (GSA-USA)

A-A-59105 - Nitric Acid, Technical.

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3. DEFINITIONS

3.1 CHEMICAL MILLING

It is the process consisting in the controlled immersion of previously prepared parts in an etchant chemical solution which removes the metal from exposed surfaces, thus producing a part with specific design configurations.

3.2 SINGLE-IMMERSION CHEMICAL MILLING

It is the chemical milling process consisting of a single immersion of parts in an etchant solution.

3.3 STEPPED CHEMICAL MILLING

It is the chemical milling process consisting of successive immersions of parts in an etchant solution.

4. REQUIREMENTS

The chemical solution is the basic element of the process, however, other factors are important, such as: proper equipment, cleaning prior to masking, suitable masking material resistant to the etch solution, scribing and removal techniques.

4.1 INITIAL CONDITIONS OF PARTS

The parts shall be previously prepared as per NE 41-012.

4.2 MATERIALS

The materials used in chemical milling process (see Table 1), shall be conform to the following basic conditions:

- a) react uniformly on the base metal;
- b) allow a good surface finish (4.8);
- c) have reasonable stability temperature range;
- d) allow chemical analytical control; and
- e) not downgrade the chemical and mechanical properties of the parts.

TABLE 1: MATERIALS USED IN CHEMICAL MILLING PROCESS

ITEM	MATERIAL	SPECIFICATION
1	NITRIC ACID (1)	A-A-59105 OR MEP 13-012
2	SODIUM HYDROXIDE (1)	MEP 13-008
3	TRIETHANOLAMINE (1)	MEP 13-032
4	SULFURIC ACID (1)	MEP 13-007
5	PERCHLOROETHYLENE (3)	ASTM D4376
6	DEOXIDIZER (1)	MEP 13-013 OR MEP 13-093
7	ALKALINE DEGREASER (1)	MEP 21-017 OR MEP 21-021
8	ADDITIVE (1)	MEP 13-087
9	DEFOAMER ADDITIVE (3)	MEP 13-091
10	MASKING COMPOUND (2)	MEP 13-074
11	ORGANIC SOLVENT (2)	MEP 13-073 OR MEP 13-097
12	KNIFE X-ACTO	(E9204659)
13	SCOTCH BRITE	(E9117480)
14	2K MASKING COMPOUND (2)	MEP 13-118

(1) Material classified as corrosive material.

(2) Material classified as flammable liquid.

(3) Material classified as toxic substance.

4.3 EQUIPMENT

4.3.1 The chemical milling tanks shall be equipped with temperature indicating and regulating devices capable of controlling the solution within $\pm 3^{\circ}\text{C}$ ($\pm 5^{\circ}\text{F}$).

4.3.2 The chemical milling tanks shall be equipped with air or mechanical agitation to minimize temperature or concentrations gradients. Agitation shall be used prior to and, where required, during chemical milling.

4.3.3 Rinsing could be accomplished by immersion (preferred), or by spray or hose. Common immersion rinse tanks could be used following deoxidizing, smut removal and chemical milling.

4.3.4 Processor qualified shall have access to precision equipment for periodically measuring surface roughness and shall meet the following requirements:

a) equipment shall measure roughness using a stylus-type averaging instrument which could be calibrated within $\pm 2\%$ of the value engraved on the reference standard;

b) the instrument shall provide a fixed (not moving) read out of the average roughness measured over a traverse of at least 5 times the cutoff distance;

c) the instrument shall utilize a diamond stylus with 5 or $10\mu\text{m}$ (0,0002 or 0,0004in) tip radius. The stylus shall contact the surface with sufficient force to permit accurate measurements, without producing a scratch which could be detected with the finger tips or nails;

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d) the equipment shall be adjustable to provide 0,25mm (0,01in) cutoff, and shall be capable of making measurements on the surfaces and 45 degrees into the fillets of chemically milled parts and/or test specimens; and

e) surface roughness requirements are specified in Ra (Roughness Average) micrometers unit (μm). RA, RHR and AA units are identical and are about 11% smaller than the obsolete RMS roughness values.

4.4 DESIGN PARAMETERS

4.4.1 GRAIN DIRECTION

As far as possible, the longest cut made for any part shall be made parallel to the grain orientation direction, that is, the longitudinal or rolling direction of the sheet or plate. Grid pattern cuts shall be laid out at 45° to the grain direction, as shown in Figure 1.

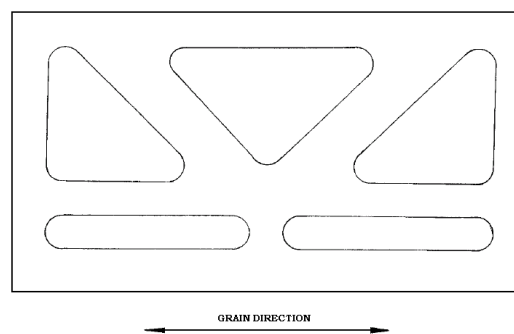


FIGURE 1: LAYOUT AS A FUNCTION OF GRAIN DIRECTION

4.4.2 TRANSITION ZONE

The transition zone is the distance from the edge of the milled cut to a point where the thickness of the part becomes constant, as illustrated in Figure 2.

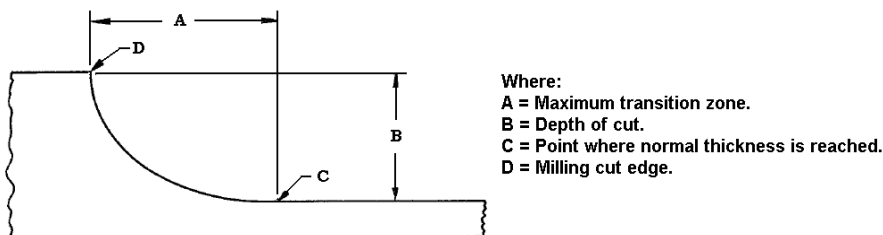
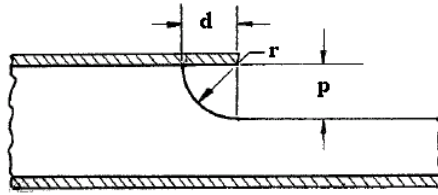


FIGURE 2: TRANSITION ZONE

4.4.3 ETCH FACTOR

The etch factor shall be determined from the ration between the etching distance (d) under the mask and the attack depth (p), as shown in Figure 3.



$$\text{ETCH RATE} = d/p$$

$$r \cong p \times f, \text{ where:}$$

r = milling radius.

f = compensation factor according to Table 2.

FIGURE 3: ETCH FACTOR

Etch factor is almost constant (around) some variation being possible to occur as a function of the material, clad coating, depth of cut, grain direction, and etchant solution composition.

TABLE 2: COMPENSATION FACTOR

ALLOY	COMPENSATION FACTOR
2024-T3/2524-T3	1,00
2024-T4/2524-T4	1,04
2024-T4 CLAD/2524-T4 CLAD/2024-T3 CLAD/2524-T3 CLAD	1,13
6061-0	1,41
6061-T4	1,30
6061-T6	1,34
6061-T651	1,34
7075-0 BARE	1,30
7075-0 CLAD	1,45
7075-T6	0,95
7075-T6 CLAD	1,60

4.4.4 DISTANCE BETWEEN CUTTING LINE OF MASK AND THE CONTOUR EDGE OF THE MILLED REGION

The calculation of the distance between the cutting lines of mask and the respective contour edges of the machined regions could vary in two ways as follows:

Ex: For a part of 2024-T3 with the characteristics of Figure 4, being 4,0mm the initial thickness of the sheet we will have the following situation:

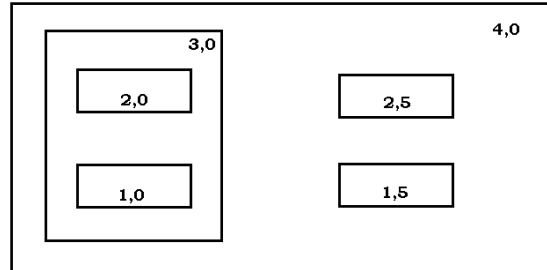


FIGURE 4: PART CHARACTERISTICS

NOTES

1) For the islands of 3,0; 2,5 and 1,5mm, the calculation of the distance between the cutting line of the mask and the contour edge (x) is equal to attack depth (p) as illustrated in Figure 5.

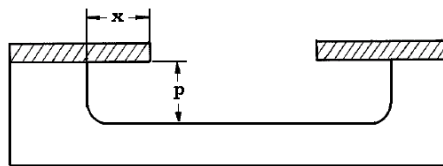


FIGURE 5: SIMPLE CHEMICAL MILLING

This happens because the contours of these islands do not present interference with any other attack, i.e., they are located in regions where the thickness around it is the initial thickness.

2) For the islands of 2,0 and 1,0mm; the calculation of “x” is obtained from the equation:

$$x_n = \sqrt{(P_n x f)^2 - p_m^2};$$

Where:

x_n = distance between cutting line of mask and the contour edge of machined region.

P_n = attack depth of island “n” in relation to the initial thickness.

P_m = attack depth of island “m”(the one which is interfering with island “n”) in relation to the initial thickness.

f = compensation factor according to Table 2.

For the case of Figure 4 ($f = 1,0$) we will have the following displacements:

$$x_{1,5} = p_{1,5} = 2,5\text{mm}$$

$$x_{2,5} = p_{2,5} = 1,5\text{mm}$$

$$x_{3,0} = p_{3,0} = 1,0\text{mm}$$

$$x_{1,0} = \sqrt{p_{1,0}^2 - p_{3,0}^2} = \sqrt{3^2 - 1^2} = 2,828\text{mm}$$

$$x_{2,0} = \sqrt{p_{2,0}^2 - p_{3,0}^2} = \sqrt{2^2 - 1^2} = 1,732\text{mm}$$

4.5 PROCESSING

4.5.1 CLEANING

Alkaline and organic cleaning and deoxidizing shall be in accordance with NE 40-012.

NOTE

Do not deoxidize or alkaline etch clean any skin quality clad surface that will not be chemically milled.

4.5.2 MASKING

After cleaning operations, the parts shall be masked according to NE 41-012 requirements.

4.5.3 SCRIBING

No scribing is needed for gauge reduction parts. For others, lay out and scribe the mask in accordance with the drawing or template, allowing for the solution undercut.

Scribe using an X-Acto knife or other sharp-pointed knife held in a near-vertical position. Cut completely through the mask. Avoid scoring the aluminum to any appreciable extent. Scribing for multiple steps could be accomplished at one time if the scribe lines are protected with a suitable line sealer.

4.5.4 RECLEANSING AND RACKING

4.5.4.1 Reclean visibly soiled surfaces by using alkaline cleaning or deoxidizing.

4.5.4.2 Rack parts for chemical milling so as to limit contact to trim areas and to minimize locations where surfaces to be milled are close to each other. Locate the parts to eliminate air pockets, to prevent settling of sludge on the parts, and to permit good circulation of solution.

4.5.5 CHEMICAL MILLING

4.5.5.1 CHEMICAL SOLUTION COMPOSITION

The etchant solution composition shall be controlled through the parameters N₁ (sodium hydroxide concentration), N₂ (dissolved aluminum concentration), N₃ (additive concentration) and triethanolamine. The concentrations for unclad aluminum alloys shall be according to Table 3.

TABLE 3: SOLUTION COMPOSITION

TEA ETCHANT	MAKEUP FOR 100 LITERS	CONTROL	CONTROL ITEM
SODIUM HYDROXIDE (N ₁)	15,5kg	120 TO 195g/l (16 TO 26oz/gal)	CHART I
SODIUM SULFIDE OR ADDITIVE (N ₃)	2,5kg (60% Na ₂ S) SOLUTION OR 2,6kg	11 TO 26g/l (1,5 TO 3,5oz/gal)	---
TRIETHANOLAMINE (N ₂)	4,3l	30 TO 60g/l (4,0 TO 8,0oz/gal)	CHART II (1)
DEFOAMER ADDITIVE (AC-930 DEFOAMER)	28ml (1fluid ounce)/ 9,3m ² (100ft ²) OF TANK SURFACE AREA	---	---
DISSOLVED ALUMINUM (N ₂)	1,0kg	5 TO 100g/l (0,67 TO 13,33oz/gal)	CHART I
ETCH RATE	---	0,03 TO 0,05mm/SIDE/min (0,0012 TO 0,002in/SIDE/min)	---
TEMPERATURE (2)	---	95 TO 105°C (203 TO 220°F)	---

(1) The triethanolamine content monitoring is just recommended but not mandatory, except for the cases mentioned in 4.8 (see observation note).

NOTE

Accumulated processing time (masking and milling) for 2024-T3XX/T4XX at processing temperatures of 93 °C (200 °F) and above is limited by the following formula:

$$(\text{temperature } ^\circ\text{C}) - 89,4 \text{ (hours)} \leq 36, \text{ or}$$

$$(\text{temperature } ^\circ\text{F}) - 193 \text{ (hours)} \leq 64$$

Ex: If a part has been subjected to a maskant cure time of 6,5h at 94°C (201°F) and 36min of chemical milling at 101°C (213°F), then:

$$(94 - 89,4) (6,5) + (101 - 89,4) (36/60) = 36$$

No further processing of this part at temperatures of 93°C (200°F) or above is allowed.

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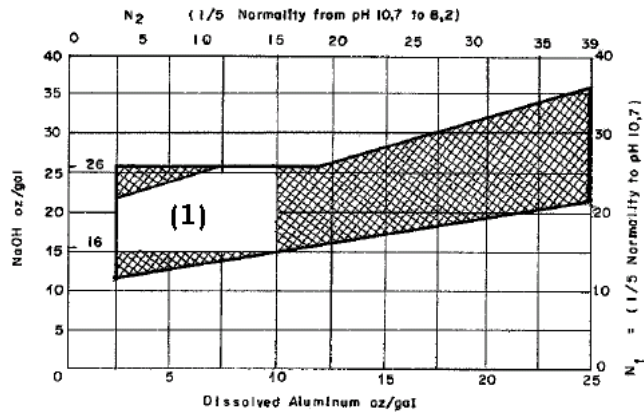
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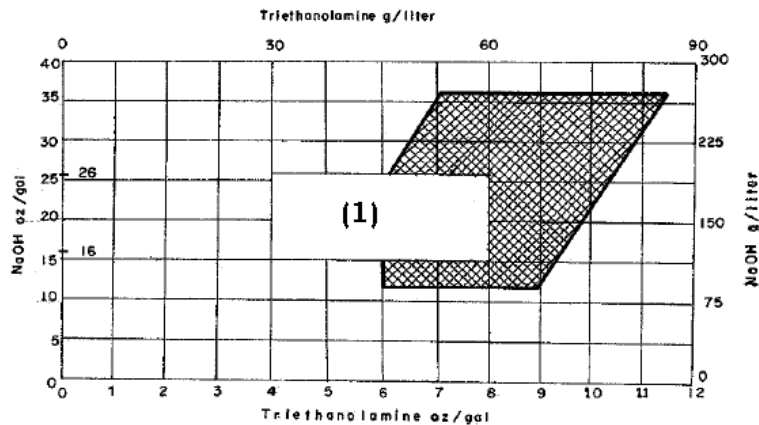


(1) Optimum ranges. See Table 3.

NOTE

N_1 and N_2 are widely used Normality Units, and could be used as an option to oz/gal for controlling sodium hydroxide and dissolved aluminum.

CHART I: CONTROL LIMITS FOR SODIUM HYDROXIDE



(1) Optimum ranges. See Table 3.

NOTE

Controls are given in oz/gal or g/l for convenience.

CHART II: CONTROL LIMITS FOR TRIETHANOLAMINE

4.5.5.2 CHEMICAL PROCEDURES

4.5.5.2.1 Check and record part thickness as required prior to chemical milling.

4.5.5.2.2 Mill the parts in solutions indicated in Table 3. If solution is operated at 93°C (200°F) or above, see 4.5.5.1, Note 2.

4.5.5.2.3 Desmut by immersion in a solution nitric acid tank. This could be followed by a hot rinse, if desired, to minimize formation of a transition zone when parts are periodically desmuted.

TABLE 4: DESMUT NITRIC ACID CONCENTRATION

MATERIALS	MAKEUP	CONTROL
HNO ₃ (53%)	350g/l	330 TO 370g/l
Fe	---	5g/l (MAX)

NOTES

1) The solution nitric acid tank used for desmutting become contaminated. Do not process parts for surface preparation for critical processes such as conversion coatings or weldings.

2) Series 7000 aluminum alloy shall be desmuted after the first 5 minutes of milling. Additional periodic desmutting is not normally required, but is recommended to produce optimum fillets, particularly at higher concentrations of dissolved aluminum. Desmutting shall be repeated for each new step started.

4.5.5.2.4. Rinse the parts in hot water at 60 to 80°C for minimum time of 1 minute. The total dissolved solids in final rinse shall be less than 1000ppm and pH 5,5 to 8,0. The parts shall be dried at the ambient temperature for the necessary time to dry or using the clean compressed air (free from humidity, oil and dust) or dried in oven at 60°C to 80°C for 10 to 30 minutes.

4.5.5.2.5 Check the depth of milling and the condition of the maskant required before completing the initial cut. Calculate milling time required for completing the remaining cut and continue.

4.5.5.2.6 Measure the depth of cut. Examine the mask and the milled surface for defects.

4.5.6 UNMASKING

Unmasking shall be made by peeling.

Peel the mask from the area to be milled using caution to avoid damaging the remaining mask. Use aluminum wool or other suitable material, if necessary, to remove all residual mask.

4.5.7 REWORK

4.5.7.1 Defects such as scratches, gouges, or pits could be removed during chemical milling by filling them with maskant, milling to a level below the defect depth, and mechanically removing the resulting island.

4.5.7.2 Maskant defects could be corrected by local repair or by stripping, cleaning, and remasking the parts.

4.5.7.3 Defects such as fillet, overhang, rough surfaces, ridging, islands, and gouges more shallow than the drawing tolerance could be removed mechanically.
Avoid damaging the surrounding metal.

4.5.8 CHEMICAL DEBURRING

The part configuration shall permit unrestricted solution and rinsing access to all surfaces.

NOTE

Do not deburr clad parts.

4.5.8.1 Thickness shall be sufficient to permit gauge reductions to remove burrs when accomplished in accordance with Table 5. Longer times are permitted, providing the gauge does not go below the minimum drawing thickness requirement.

4.5.8.2 Parts that contain holes, threads, polished surfaces, or other dimensional features where metal removal would be objectionable shall be masked before deburring.

4.5.8.3 Parts with welds or dissimilar metal inserts, which could be subject to preferential attack, shall be masked before deburring.

**TABLE 5: METAL REMOVAL RATE AND IMMERSION TIMES FOR
CHEMICAL DEBURRING**

METAL THICKNESS mm (in)	METAL TO BE REMOVED PER SIDE mm (in)	IMMERSION TIME (minutes) 0,0254mm/side/min (0,001in/side/min)
MINOR BURRS - METAL THICKNESS TO 1,016 (0,040)	0,0127 TO 0,0381 (0,0005 TO 0,0015)	0,5 TO 1,5
1,041 TO 2,54 (0,041 TO 0,100)	0,0381 TO 0,0762 (0,0015 TO 0,0030)	1,5 TO 3,0
OVER 2,54 (0,100)	0,0762 TO 0,1143 (0,0030 TO 0,0045)	3,0 TO 4,5

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4.6 APPEARANCE

The parts shall be finished, clean and free from processing residues that could cause subsequent discoloration and corrosion. The finished surface (after etching) shall be free from pits, dents, scratches, and other evidence of irregular etchant action.

Scratches which could be noticeably felt with finger nail shall be cause for rejection.

Light scuff marks and minor abrasive marks are acceptable.

Unevenness of edges due to masking imperfections shall not be considered as pits. Pitting on trim areas which will subsequently be removed from the parts shall not be considered objectionable.

NOTE

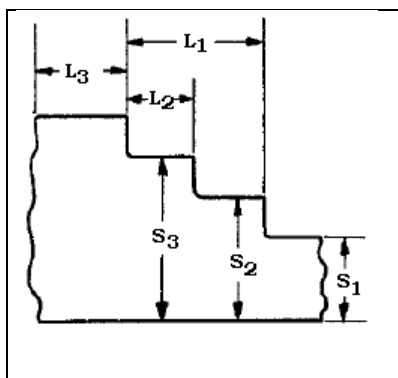
The occurrence of pits in places other than chemical milled surfaces shall be submitted to the Materials Engineering approval.

The pits in exterior surface of skin are not acceptable.

4.7 DIMENSIONAL TOLERANCES

Tolerances for the smallest thickness and the intermediate of the part are indicate in Table 6.

TABLE 6: DIMENSIONAL TOLERANCES (1)

	THICKNESS "S"	TOLERANCE "s" (mm)	WIDTH "L"	TOLERANCE "L" (mm)
	S ₁	-0,10 + 0,15	L ₁	±1,3
	S ₂	±0,15	L ₂	±1,3
	S ₃		L ₃	±0,8

(1) If any part requires tolerance greater or smaller than those specified, they shall be specified on drawing.

NOTE

The calculation of the material removal during the chemical milling shall be based on the actual part thickness and not in the nominal thickness.

4.8 SURFACE ROUGHNESS

Unless otherwise specified on the production drawings or in the contract, the surface roughness (Ra) of the parts shall not be in excess of 3,2µm, as defined in NE 03-004.

NOTE

If the roughness is >3,2µm, it shall be analyzed all components of the attack solution, mainly the triethanolamine (TEA) content.

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4.9 RAISED AREAS

Raised areas or small islands, created by residual masking or insufficient etching action, shall be removed by mechanical methods such as filing, sanding, or grinding. Removal shall neither adversely affect mechanical properties and the specified dimensional tolerances nor increase surface roughness or susceptibility to corrosion. See Figure 8-F.

4.10 LINE DEFINITION AND EDGE WAVINESS

4.10.1 Line definition is the extent to which the actual etched straight line deviates from the ideal (design) etch line.

Perfect masking shall produce a clean straight edge. However, there could be deviation due to cutting (see NE 41-012), imperfection in masking, or mistakes in determining the proper etch factor (see 4.4.3).

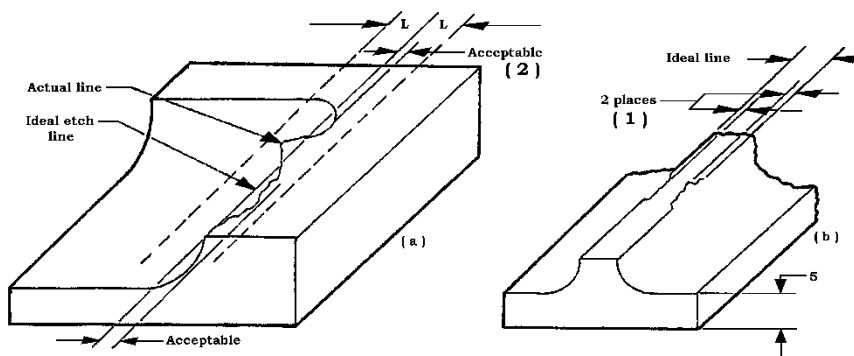
The edge of unmilled area could be uneven and is acceptable provided the total variation from the ideal line does not exceed the specified values in Table 6.

See Figure 6.

Masking imperfections at the edges and fillets shall not deviate from the ideal etch line by more than the amount show in Table 7. See Figure 6.

TABLE 7: LINE DEFINITION ALLOWANCE

DEPTH OF CUT (mm)	MAXIMUM DEVIATION (mm)
0 – 2,54	1,02
2,55 – 5,08	1,65
5,09 – 7,62	2,03



- (1) Not acceptable, unless rib width meets tolerance specified on engineering drawing.
 (2) Maximum deviation acceptable according to Table 6.

FIGURE 6: TOLERANCE FOR LINE DEFINTION

4.10.2 The edge waviness and scribe line mismatches are acceptable provided that:

- a) the depth of the wave or the depth of mismatch (p_o) does not exceed the depth of cut (p) or the limits specified in Table 6;
- b) the waviness does not extend into flat chemical milled area; and
- c) $r_1 \geq p_o$ and $r_2 \geq 2 p_o$. See Figure 7.

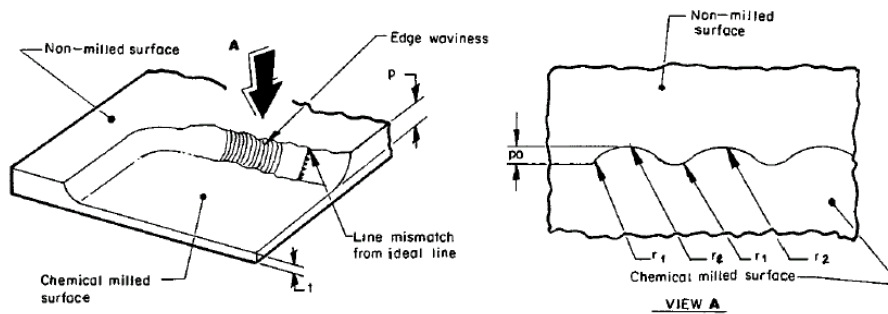


FIGURE 7: EDGE WAVINESS AND LINE MISMATCH FROM IDEAL LINE

4.11 FILLET RADII

Fillet radii shall not be specified as they are inherent and fixed by the process itself. Radii will vary with material, heat treatment, depth of cut, and grain direction of the part material.

Roughly, the radius value could be considered as approximately equal to the depth of cut (see Figure 3).

Reworking in transition zones (see 4.4.2), is permissible to obtain conditions for compliance with the requirements of the applicable drawing. Grinding devices as well as abrasive papers could be used for such reworking.

4.12 SHARP EDGES

Chemically milled parts shall not present any sharp edges, corners, or projections to ensure acceptance.

All sharp edges from chemical milling shall be broken to a radius between 0,13 to 0,76mm.

NOTE

All burrs, sharp corners and sharp edges on parts shall be removed prior to chemical milling, unless otherwise specified in drawings.

4.13 DEFECTS LIABLE TO ARISE FROM CHEMICAL MILLING

Most frequent defects in chemically milled parts, their causes, repair possibilities, and acceptability are presented herein.

These defects are schematically illustrated in Figure 8.

NOTE

Care shall be taken on rework of alclad surfaces so that not more than 15% of the clad is removed.

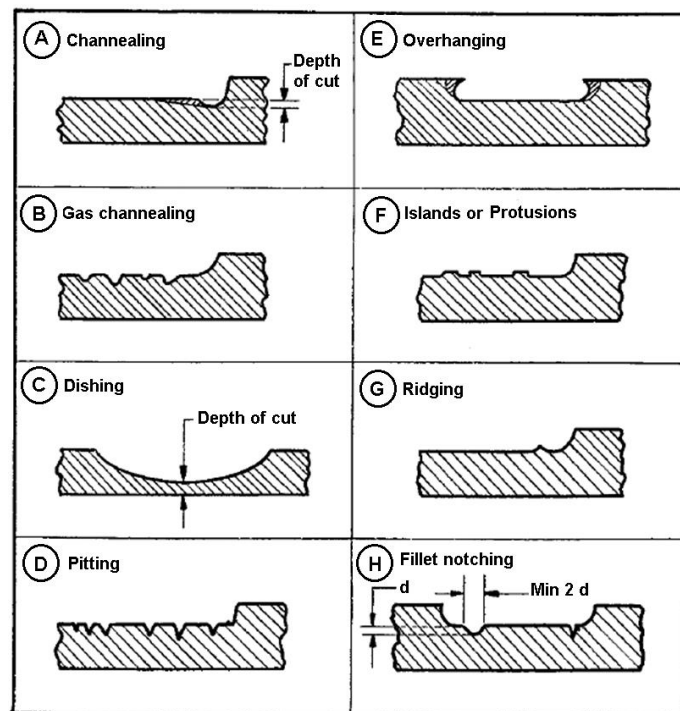


FIGURE 8: DEFECTS ON CHEMICALLY MILLED PARTS

4.13.1 CHANNELING

This defect as illustrated in Figure 8-A, is the formation of a channel or groove at the base of the fillet mainly due to excessive concentrations of dissolved metal ($N_2 \geq 90$).

This condition is undesirable but is acceptable provided it could be corrected by mechanically blending the undercut, as shown in Figure 8-A (opposite hatching), and all required dimensions be within tolerances specified in 4.7. See Figure 9.

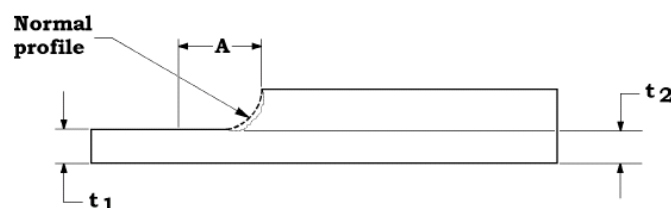


FIGURE 9: CHANNELING

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Acceptable if:

- a) t_1 and t_2 are within the tolerances specified in 4.7;
- b) the transition distance “A” is equal to or greater than 10 times the depth of channeling (t_1 and t_2); and
- c) the depth of channeling (t_1 and t_2) is not greater than 0,25mm or 20% of t_1 thickness, whichever is less.

4.13.2 GAS CHANNELING

This surface condition, as illustrated in Figure 8-B, is the result of bubbles of hydrogen or other gases ascending to the surface of the etch bath. The resultant formation of vertical grooves or channels in the etched areas shall be cause for rejection unless corrected by a process approved by Materials Engineering and all dimensions are within tolerances specified in 4.7.

Gas channeling is mainly ascribable to an incorrect positioning of parts in bath.

4.13.3 DISHING

Dishing, as illustrated in Figure 8-C, is due to an excessive concentration of N_1 or N_2 , or both as well as to improper agitation or temperature control of the etchant.

Dishing consists in thinning of the central portion of the metal remaining after milling. It is acceptable provided all required dimensions are within the specified tolerances and, chiefly, the thickness of the metal at the thinned central portion is not less than the specified minimum thickness.

4.13.4 PITTING

This defect, as illustrated in Figure 8-D, appears as pits or depressions formed in the etched area mainly due to excessive concentrations of sodium hydroxide solution ($N_1 \geq 195\text{g/l}$).

Pits formed by chemical milling shall not exceed 1,5mm (0,06in) in diameter and 0,025mm (0,001in) in depth or 5% of material thickness, whichever is less, with the following limitations:

- a) pits shall not be acceptable if located in radii of structural parts, such as: spar, shapes, beams, or other structural members (see Figure 10);
- b) pits shall be at least twice their diameter away from part corners or at least twice their diameter away from fastener hole;
- c) pits shall be limited to two per 645mm^2 and to 5% of overall part area; and
- d) pits shall not pass through clad of aluminum alloy parts.

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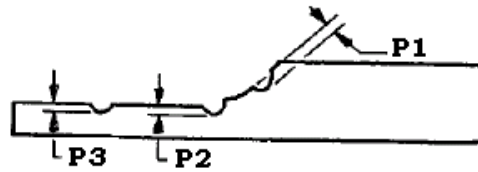
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Pitting out of the mentioned criteria (“a” to “d”) above shall be cause for rejection, unless the condition could be remedied by a process approved by Materials Engineering within the tolerances specified in 4.7.



P₁ and P₂ are not acceptable.

P₃ is acceptable if the depth does not exceed 0,025mm and meet the requirements of “b”, “c” and “d”.

FIGURE 10: LOCATION OF PITS

4.13.5 OVERHANGING

This defect, illustrated in Figure 8-E, consists in the presence of overhanging metal over the etched area and shall be cause for rejection, unless the overhang (indicated in Figure 8-E by opposite hatching) is cleared off by a process approved by Materials Engineering and the required dimensions remain within the tolerances specified in 4.7.

4.13.6 ISLANDS OR PROTRUSIONS

This defect, illustrated in Figure 8-F, is the result of unwanted masking action of dirt or particles of maskant which remain on the surface to be etched. It appears as high spots within the etched area.

This surface condition shall be cause for rejection unless the islands are removed by a process approved by Materials Engineering and all dimensions are within tolerances specified in 4.7.

4.13.7 RIDGING

This defect, illustrated in Figure 8-G, is the formation of a raised ridge at the base of the fillet, due to excessive concentrations of dissolved metal ($N_2 \geq 90\text{g/l}$) or to improper positioning for parts in etch bath.

This condition is undesirable and cause for rejection, unless the ridge is removed by process approved by Materials Engineering and all required dimensions remain within tolerances specified in 4.7.

The presence of ridge in the fillet is acceptable provided that the radius “r” is greater than 80% of the depth of cut (p). See Figure 11.

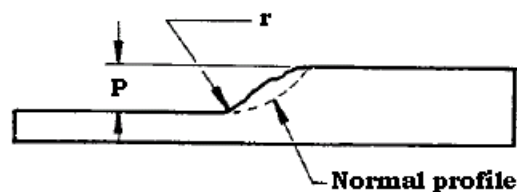


FIGURE 11: RIDGE IN FILLET (MILLED)

4.13.8 FILLET NOTCHING

This defect, illustrated in Figure 8-H, is the result of scribing too deeply (see NE 41-012). Notching is acceptable when its configuration is as shown on left side of Figure 8-H, that is, when the depression width is at least twice the notch depth, provided the thickness of the metal part at the root of notch is within the tolerances specified in 4.7; otherwise, as shown on right side of Figure 8-H, the part shall be rejected.

4.14 MECHANICAL PROPERTIES

The mechanical properties of the metal parts, as detailed in the applicable material specifications, shall not be downgraded or decreased by chemical milling.

4.15 CORROSION RESISTANCE PROPERTIES

The parts shall be finished clean and free from processing residues that could cause subsequent corrosion and discoloration. The material used for chemical milling shall not result in any attack boundaries or weldments, pitting or other metal degradation.

Parts with pitted surfaces or showing intergranular attack shall be rejected.

Chemical milling shall not produce intergranular attack to a depth greater than 0,005mm (5µm) or pitting to a depth greater than 0,013mm (13µm).

5. QUALITY ASSURANCE PROVISIONS

5.1 PROCESS CONTROL

5.1.1 INSPECTION OF PARTS

For Quality Assurance, all parts chemically milled as per this Standard shall be inspected for appearance, dimensional tolerances, surface roughness, line definition, fillet radii, and sharp edges, according to 4.6 to 4.13.

In cases of stepped milling (see 4.4.4), parts shall be inspected after each operation.

5.1.2 DEFECTS

Chemically milled parts presenting the defects described in 4.13 shall be inspected according to criteria therein.

5.1.2.1 PITTING

Moderate pitting resulting from damage or imperfection in the mask (not considered as pits) is acceptable provided:

- a) pit does not exceed 1,5mm diameter;
 - b) pit depth (on aircraft inner surfaces) does not exceed 10% of material thickness in pitting area;
 - c) pit depth (on aircraft outer surfaces) does not exceed 10% of material thickness in pitting area or the maximum value of 0,25mm;
 - d) pit distance from transition zones (see 4.4.2), recesses, and any other sudden changes in part contour or thickness is 6,4mm minimum;
 - e) pit distance from fastener attaching points is 13mm minimum; and
 - f) pits are at least 13mm apart, are not aligned, and do not present elongation.
- In case of parts milled on both sides simultaneously, pitting on both sides is considered as being on one side only for the purposes of determining spacing, continuity, and elongations.

NOTE

Pits meeting all the items above shall be smoothed away as possible so that the ratio between minimum diameter and maximum depth is equal to or greater than 10:1. This operation shall produce a roughness surface equivalent to or better than that in adjacent areas (see 4.8) and shall not detrimentally affect part mechanical properties or increase its susceptibility to corrosion.

5.1.3 INTERGRANULAR ATTACK

5.1.3.1 SPECIMEN REQUIREMENTS

The below requirements shall be followed:

- a) for all aluminum and aluminum alloys, except aluminum-lithium, cut an extruded section of 7075-T6511 or 7175-T6511 such that the end grain is exposed. Prolongations or trim allowance could be used in lieu separate coupons;
- b) polish the surface(s) to be examined to RHR32 or finer surface finish.

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5.1.3.2 PROCEDURES

The steps below shall be followed:

- immerse the specimens into the TEA etchant solution (see Table 3) during 30 minutes or for the time necessary to remove 0,0254mm (0,0010in) of metal per surface;
- afterwards, rinse in distilled water and allow to dry at ambient temperature;
- cross section the specimen normal to the surface being evaluated and parallel to the grain;
- prepare adequately the specimens for mettalographic analysis and examine the specimens with microscope up to 100X magnification; and
- measure intergranular attack and end grain pitting for compliance with the item 4.15. See Figure 12 for example.

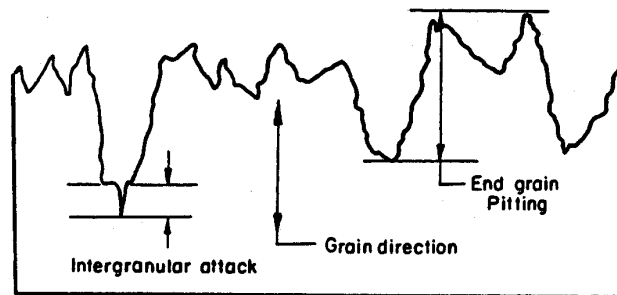


FIGURE 12: EXAMPLE OF INTERGRANULAR ATTACK AND
END GRAIN PITTING

5.1.4 CHEMICAL MILLING TEST SPECIMEN

The qualification specimens shall be chemically milled to the dimensions shown in Figure 13. Specimens shall be aluminum alloy 2024-T3 or T4 without clad.

No mechanical rework is permitted during or after processing of qualification specimens. The surface roughness shall be according to the requirements of item 4.8 and the thickness shall be according to the requirements of Table 6.

5.1.5 SOLUTION TEMPERATURE GRADIENT

Solution temperature gradient in the chemical milling solution shall be $\pm 3^{\circ}\text{C}$ ($\pm 5^{\circ}\text{F}$), based on temperature uniformity to be verified according to NE 40-077 in the Process Qualification and when to evidence irregularity or discrepancy in the process that compromise directly the solution temperature uniformity.

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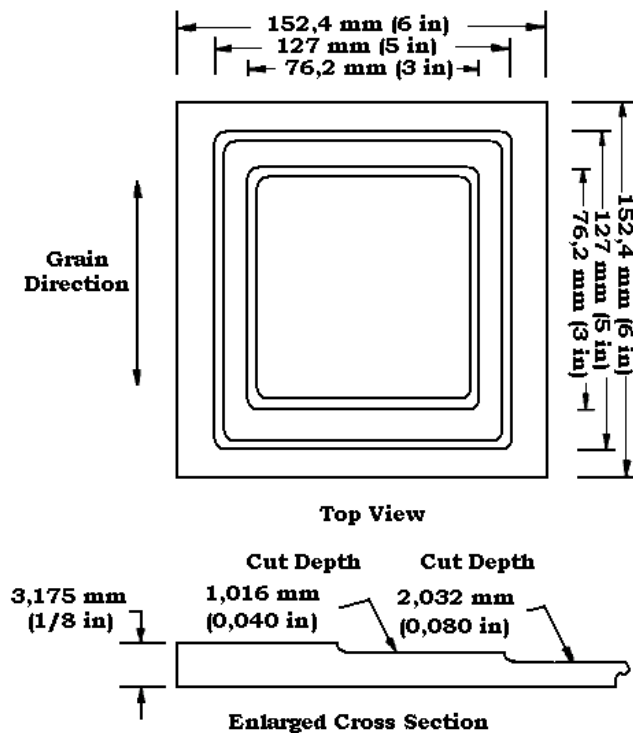


FIGURE 13: CHEMICAL MILLING TEST SPECIMEN

6. GENERAL NOTES

6.1 TREATMENT OF RESIDUES

Before adjusting the etchant solution (4.5.8), the etchant tank discharges a large amount of pollutants (such as sodium hydroxide, metallic particles, sulfides and additives), into sewers, therefore an effective treatment for neutralization of these residues shall be carried out.

The NaOH shall be neutralized by means of preferable sulfuric acid whereas the sulfides, by commercial sodium hypochlorite or chlorine in bottles.

6.2 EXAMPLE OF ETCHING TIME CALCULATION

The etching time is equal to the relationship between the thickness of material to be removed (depth of cut) and the etch rate. The etch rate, although almost constant around 55µm/s (side x minute) shall be calculated with a greater accuracy.

An example of this calculation is given as follows:

Suppose that a 1,90mm deep region is to be milled in a 2024-T4 non-clad sheet 3,18mm (0,125in) thick. Based on his experience, the process operator knows that the etch rate varies around 50µm/s (side x minute).

Since only one of the part is exposed, a preliminary calculation indicates that the etch rate shall be (1900/50 = 38 minutes). At first, the operator shall measure the part thickness by using a micrometer (3,18mm).

Next, he shall expose the part to chemical etching for a time approximately equal to half the time previously calculated (= 19 minutes), wash the part, and measure the thickness of region etched (2,07mm), by using a micrometer calculating then the depth of cut ($3,18 - 2,07 = 1,11\text{mm}$), and the actual etch rate ($= 1,11\text{mm}/19\text{min} = 1110\mu\text{m}/19\text{ min} = 58\mu\text{m}/\text{minute}$).

Then, the operator calculates the etch time necessary for milling the remaining thickness ($790\mu\text{m}/58\mu\text{m}/\text{minute} = 13,6\text{ minutes}$).

Therefore, overall immersion time for this case shall be $19 + 13,6 = 32,6\text{ minutes}$.

6.3 PROCEDURES

At Embraer, the procedures for carrying out the chemical milling process of aluminum and its alloys are specified in doc.emb 005901.

NOTE

The procedures indicated in doc.emb are adapted to Embraer manufacturing conditions, therefore the partners could use any procedures that comply with the requirements of this Standard.

6.4 MECHANICAL MILLING

Mechanical milling on sheet parts could be considered, applying specific tolerances criteria according to NE 03-071. NI 1531 and NI 624 shall be called on drawings.

6.5 DRAWING REPRESENTATION

The representation on drawings, of the area to be chemically milled, shall be according to specific Drafting Standard Manual of each program.

6.6 DRAWING CALLOUT

This Standard shall be called out on production drawings through following note:

NI 721 - Chemical milling according to NE 41-013 ("NI 721 – Usinar quimicamente conforme NE 41-013").

NOTE

On CAD drawings where the countour(s) of the surface(s) subject chemical milling is closed compound curve(s), the following Engineering note shall be also indicated:

NI 749 - Total area of surface(s) for chemical milling $\langle * \rangle \text{m}^2$, total perimeter of surface(s) for chemical milling $\langle ** \rangle \text{m}$.

* Indicate nominal value of total area with three decimal digits (X,XXX).

** Indicate nominal value of total perimeter with three decimal digits (X,XXX).

The total area and total perimeter are represented by the addition of the areas plus the perimeters of the punctual surfaces with closed contours subject to chemical milling one a same part.

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6.7 BIBLIOGRAPHIC REFERENCES

MIL-C-81769 - Chemical Milling of Metals - Specification for.

DPS9.481 - Chemical Milling Requirements for Aluminum Alloys.

Report 000EBD005 – "Análise do Processo de Usinagem Mecânica em Chapas de Alumínio".

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